

Final Report Cover Sheet

Project Team Number: C1

Public repository copy. Student numbers and family names have been redacted.

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Executive Summary

Plastic waste on university campuses and in public spaces is often improperly sorted. As the volume of plastic waste generated worldwide increases, manual sorting is becoming less suitable for managing this problem at scale.² Autonomous robots offer a potential solution by providing faster, more accurate, and cheaper plastic waste sorting. Accordingly, this project aims to develop an autonomous prototype robot that sorts plastic waste, represented by coloured balls, into correct categories to explore how automation can improve recycling efficiency and sustainability. The robot will be required to sort red, green, and blue balls, representing HDPE, PET, and PP plastic types respectively, while avoiding yellow balls, which represent non-recyclable material, within a defined arena in an operator-specified order by navigating to the collection zone to collect balls and returning to its base to deposit them. The following report presents the design team's understanding of the design problem, outlines the robot's technical specifications, explains its design evolution, and reflects on and evaluates the robot's performance, with suggestions for future work.

The design for the waste sorting robot is a self-contained WALL-E-inspired system. The robot is based around a dual-level PLA chassis with an outer shell and moves using two front wheels and a rear caster wheel. It is controlled by an Arduino Uno microcontroller and powered by two 9 V alkaline batteries and four AAA batteries. The robot also features several sensors to detect balls and obstacles. To collect balls, it uses a curved, servo-actuated clamp that closes around each ball.

During development, the design team focused on keeping the design simple, reliable, and easy to modify. As such, notable design features include an LCD for simple and clear user feedback, a durable collision-resistant PLA chassis, and a subsystem breakdown that allows each part of the robot to be developed simultaneously and easily adjusted. However, the final design of the robot is limited as it is specialised for ball collection in a controlled arena and would require significant modification for real-world plastic waste sorting.

Overall, the design successfully met the project's performance objectives in a cost-effective manner by accurately detecting and sorting different coloured balls. The project also provided valuable insight into the effectiveness and potential of autonomous waste-sorting robots.

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1 Final Design Implementation

1.1 Problem and Objectives

Plastic waste on university campuses and in public spaces is often improperly sorted, leading to contamination of recycling streams and reduced effectiveness. Indeed, a 2024 Cleanaway report found that 40% of Australians do not separate soft plastics from hard plastics before recycling.¹ As the volume of plastic waste generated worldwide increases, manual sorting, which is labour-intensive and prone to error, is becoming less suitable for managing this problem at scale. This project aims to address this issue by developing an autonomous prototype waste sorting robot to explore how automation can improve recycling efficiency and sustainability. The robot must operate within a defined arena and sort red, green, and blue balls, representing HDPE, PET, and PP plastic types respectively, while avoiding yellow balls, which represent non-recyclable material. It must also collect and sort the balls in an operator-specified order. There were a number of restrictions on the design. These are: the robot must be self-contained, with maximum dimensions of 200 mm x 200 mm x 200 mm; the robot must use no more than 50 AUD in additional materials beyond the provided kit; the project must be completed and ready for testing by 23 April 2026.

1.2 Concept Overview

The final design is an autonomous, WALL-E-inspired waste sorting robot. At the start of operation, the robot logs its base colour. It then turns and navigates to the central collection zone using two front drive wheels and a rear caster wheel. Once there, it uses a PixyCam 2.1 and other sensors to locate the target coloured ball while avoiding obstacles. It then collects the target ball using a front-mounted clamp mechanism and returns to its logged base colour, where the ball is deposited. The overall design of the robot is shown in Figure 1.

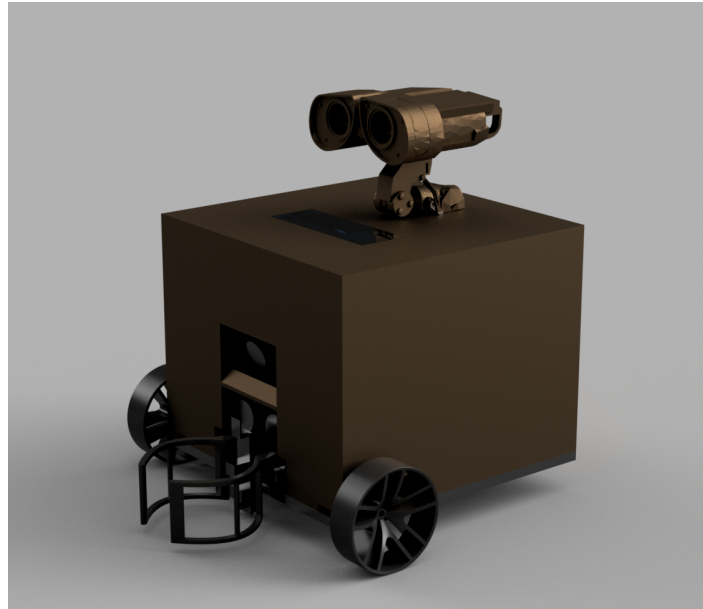


Figure 1: 3D render of final waste sorting robot design

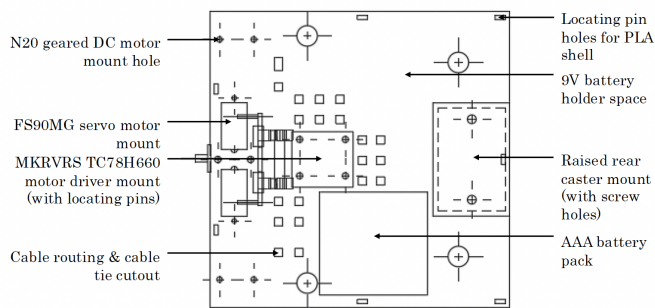
During development, the design approach focused on keeping the design of the robot simple, reliable, and easy to modify. As such, the robot was divided into four distinct subsystems. This modular breakdown helps simplify system integration and allows different parts of the robot to be modified without redesigning the overall system. The four subsystems are: the Control Subsystem, the Vision

and Detection Subsystem, the Chassis and Navigation Subsystem, and the Ball Handling Subsystem. Further detail about each subsystem and their interactions is explained in Section 1.4, Subsystems and Integration.

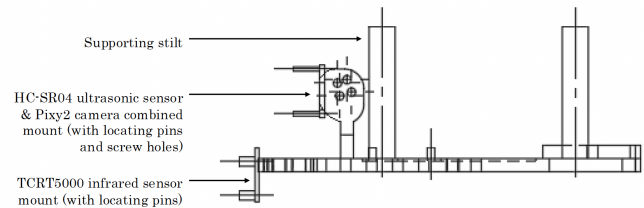
1.3 Detailed Design

1.3.1 Mechanical Design

The chassis is 3D-printed from PLA and split into two levels, each using screw holes, cutouts, and locating pins to mount the components. The bottom level is 146.5 mm x 145 mm and contains the following components: two 9 V alkaline batteries, four AAA batteries, two N20 geared DC motors, the MKRVRS motor driver, two FS90MG servo motors with 3D-printed clamp arms, two front wheels with rubber tyres, a rear caster wheel, a PixyCam 2.1, an HC-SR04 ultrasonic sensor, and a TCRT5000 infrared sensor. The layout of the bottom level can be seen in Figure 2.



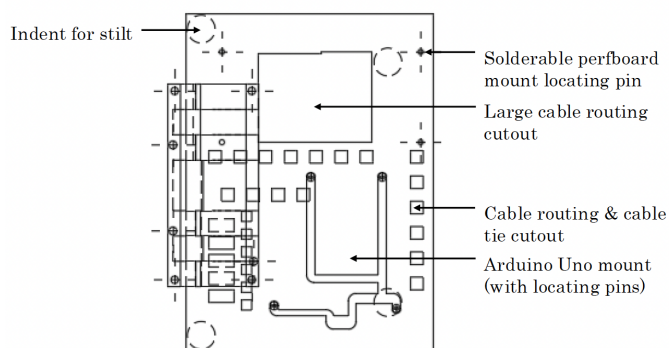
(a) Top view of the chassis' bottom level



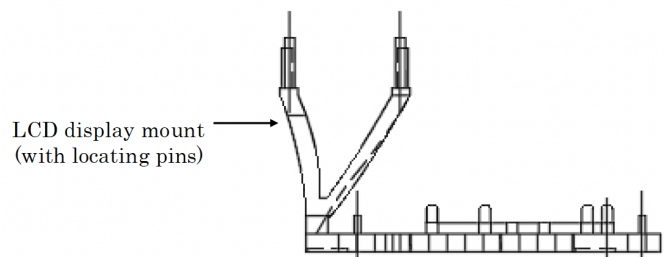
(b) Side view of the chassis' bottom level.

Figure 2: Technical drawings of the bottom level of the robot chassis

The top level is 98 mm x 133 mm and contains the robot's logic: an Arduino Uno, an Arduino shield, a perfboard, and an LCD display. The layout of the top level is shown in Figure 3.



(a) Top view of the chassis' top level



(b) Side view of the chassis' top level

Figure 3: Technical drawings of the top level of the robot chassis

The overall layout of the chassis is shown in Figure 4.

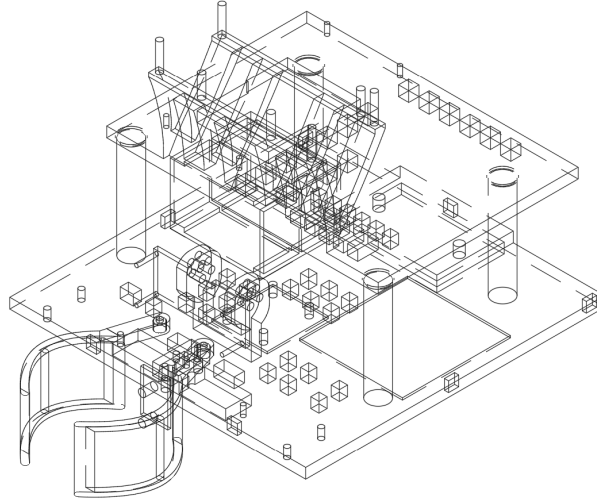
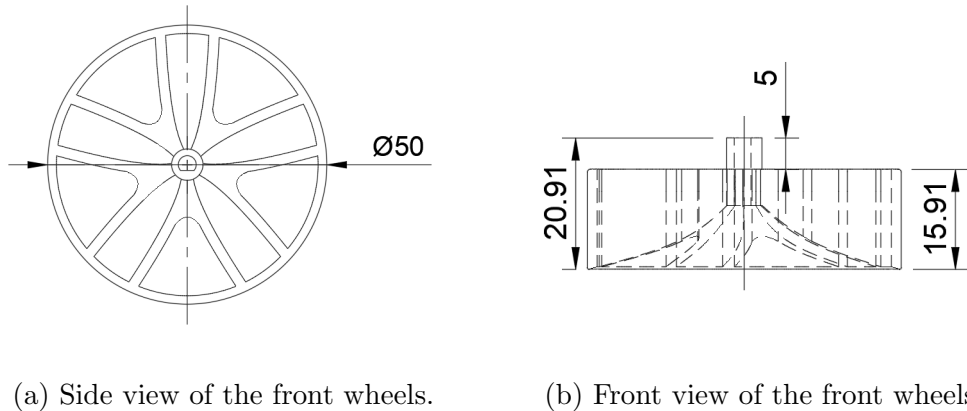


Figure 4: Technical drawing of the robot chassis without outer shell, isometric view

Cable management is handled through small cutouts on both levels which are used to route wires and anchor cable ties, keeping the wiring clean and easy to trace. A larger cutout also sits directly beneath the perfboard, linking the two levels. The PixyCam and HC-SR04 ultrasonic sensor mount is integrated into the baseplate, as shown in Figure 9a. The HC-SR04 slots into a small plate using locating pins that align with its screw holes, with space left behind for cable clearance. The PixyCam attaches to a separate small plate using its own screw holes. This plate is then bolted to the upper part of the mount using one of two hole positions, setting the camera at either a 10° or 20° downward angle.

The drive system uses two front wheels powered by N20 motors and a rear caster wheel, allowing the robot to turn on the spot. The front wheels were designed in-house with a Y-spoke pattern, shown in Figure 5.



(a) Side view of the front wheels.

(b) Front view of the front wheels.

Figure 5: Technical drawings of the front two 3D printed wheels

The overall layout of the chassis is shown in Figure 4. The clamping mechanism is positioned at the

front of the robot and is intentionally simple. Two servos sit in cutouts on the baseplate, each connected to a clamp arm. The servos rotate in opposite directions to close around the ball. Between the servos, the TCRT5000 IR sensor is mounted on two small locating pins that pass through its screw holes. The clamp mechanism is shown in Figure 6.

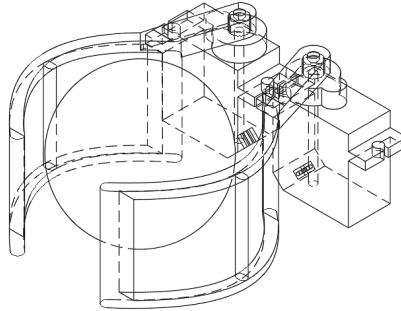


Figure 6: Technical drawing of the ball collection clamp mechanism, isometric view

Sitting over the dual-level base is a removable PLA shell that attaches using locating pin holes around the edge of the bottom plate, allowing it to be easily removed and reattached. The shell includes cutouts for the LCD display, PixyCam, HC-SR04, and TCRT5000, and is finished with rust-coloured spray paint and a WALL-E head mounted on top to match the robot’s intended aesthetic.

1.3.2 Electrical Design

The Arduino Uno microcontroller functions as the main control unit for the robot.³ A shield is fitted over the Arduino to provide reliable electrical connections. As described in Section 1.3.1, Mechanical Design, the Arduino is mounted on the top level of the chassis. A solderable perfboard serves as the main connection pathway between the two levels, distributing 5 V and GND through power rails formed on each level. Any additional cables are routed through cutouts in the baseplate.

The circuit is powered by two 9 V alkaline batteries and four 1.5 V AAA alkaline batteries. The first 9 V battery powers the Arduino via the power jack, with a clip and switch soldered to the power jack for connection. The second 9 V battery supplies the MKRVRS motor driver, while the four AAA batteries power the two FS90MG servo motors. The motor driver’s regulated 5 V output is also connected to the shared 5 V and GND rails³ to provide additional current to other components if needed. A power switch is also included in the circuit to allow the robot to be easily turned off and on.

To smooth power fluctuations, two 1000 μF capacitors are used. One capacitor is placed across the main power rails, and the other is placed across the terminals of the 9 V battery supplying the Arduino. All components share a common ground with the Arduino to provide a 0 V reference, ensuring signals can be transmitted reliably across the system.³

The remaining component connections are shown in Figure 7. The perfboard and power switch are not included in Figure 7 to maintain clarity.

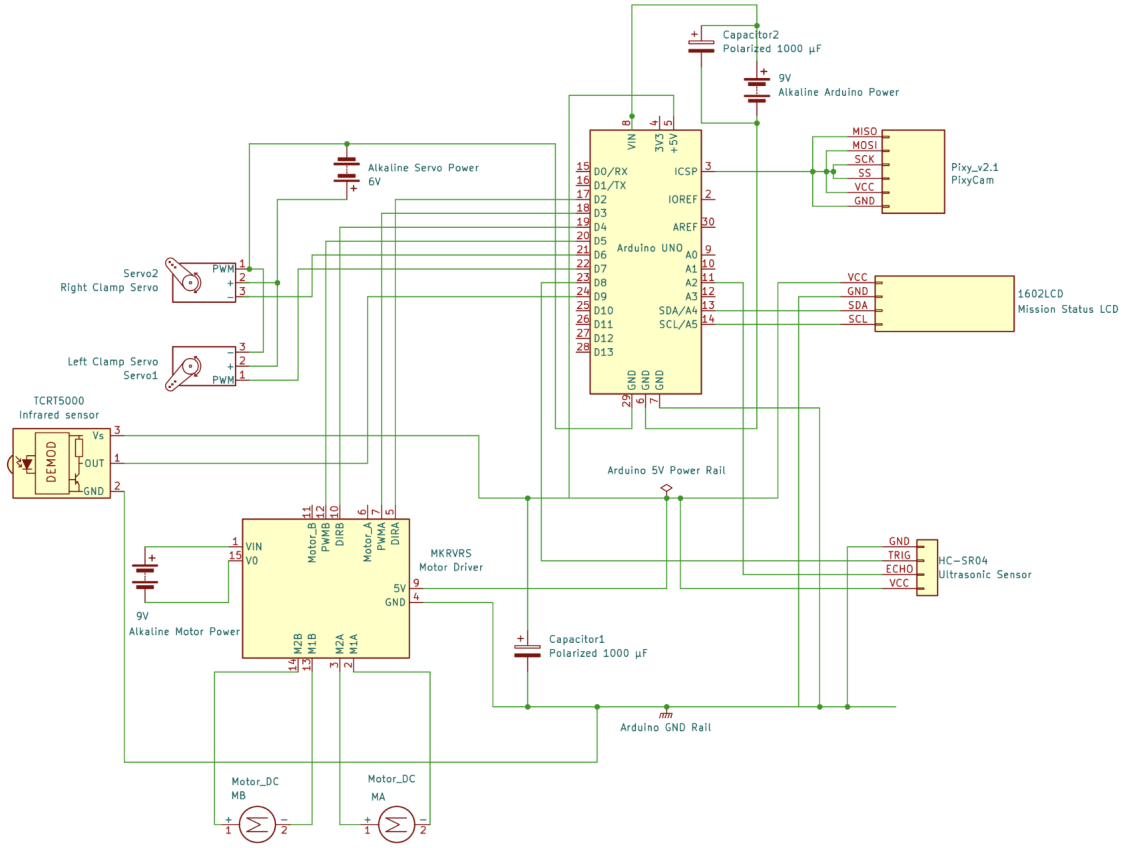


Figure 7: Circuit Diagram of the robot, without perfboard and SPDT switch.

1.3.3 Software Design

The software is implemented on the Arduino Uno using a modular structure with clear task separation.³ The `setup()` routine initializes sensors and actuators, logs the base signature at startup, and establishes a known mechanical state. The `loop()` routine then executes repeated colour cycles in the required order. Each cycle calls dedicated functions for search, centring, chase, clamp actuation, return-to-base, and deposit, so motion control and decision logic remain consistent across all three target colours. Operational reliability is ensured through rule-based checks and feedback at each stage. PixyCam 2.1 block position and size are used for steering and approach control, the IR sensor confirms capture before transition to return mode, and ultrasonic readings support obstacle handling during normal navigation.^{5,7,10} Timeout limits, confirmation checks, and orientation state tracking are also applied to reduce false transitions and prevent repeated turning errors, which helps the robot recover from temporary loss of visual targets and continue the mission sequence. See Section 1.4.5, Subsystem Integration for more details.

1.4 Subsystems and Integration

1.4.1 Control Subsystem

The Control Subsystem manages the overall operation of the robot.³ It consists of an Arduino Uno microcontroller and an LCD display, which provides simple user feedback.

The subsystem operates by the Arduino receiving sensor data from the Vision and Detection Subsystem. Specifically, the PixyCam 2.1 detects coloured balls and the robot's base, the HC-SR04 ultrasonic sensor

detects obstacles, and the TCRT5000 infrared sensor detects when a ball is in the clamp. The Arduino processes this data using conditional logic in the code to determine the commands sent to the Chassis and Navigation Subsystem or the Ball Handling Subsystem. This loop continuously repeats by reading sensor inputs, evaluating conditions, and sending commands. These commands are sent using PWM signals.⁶ For the MKRVRS motor driver, the PWM duty cycle controls motor speed, thereby controlling the movement of the robot.⁶ For the FS90MG servo motors, the PWM pulse width controls the rotation angle, thereby opening and closing the clamp.¹²

1.4.2 Vision and Detection Subsystem

The Vision and Detection Subsystem is responsible for gathering information about the robot's surroundings and sending this sensor data to the Control Subsystem. It consists of a PixyCam 2.1, a HC-SR04 ultrasonic sensor, and a TCRT5000 infrared sensor.

The PixyCam uses trained colour signatures in Colour Connected Components mode to detect red, green, and blue balls, as well as base locations.^{4,5} Signatures 1 to 3 identify red, green, and blue balls respectively, while signatures 4 to 7 identify the four corner bases. At startup, the robot logs its base signature and the corresponding signature size in pixels, enabling accurate return navigation and ball deposit. The HC-SR04 supports navigation by measuring distance to obstacles using 40 kHz ultrasonic pulses.⁷⁻⁹ To address noise-induced spikes in distance readings, a median filter is applied in the code to batches of five measurements to remove outliers and output the median value.⁸ In addition, a time-jitter method randomises trigger timing, reducing the likelihood of interference from other ultrasonic sensors operating in the arena. The TCRT5000 uses infrared light to detect balls within the clamp.^{10,11} When a ball enters the clamp area, reflected infrared light causes the sensor to output a LOW signal, which the Arduino reads and uses to send a command to close the clamp.^{10,11} When no ball is present, it outputs HIGH.

1.4.3 Chassis and Navigation Subsystem

The Chassis and Navigation Subsystem is responsible for providing physical housing for all components and moving the robot throughout the arena. It consists of a PLA chassis, an MKRVRS motor driver, two N20 geared DC motors, two front wheels with rubber tyres, and one rear ball caster wheel. The structure and layout of the chassis are explained in Section 1.3.1, Mechanical Design.

The subsystem operates by the MKRVRS motor driver receiving movement commands from the Arduino Uno in the Control Subsystem in the form of PWM signals. The PWM duty cycle controls the average voltage applied to the DC motors, thereby controlling wheel speed through the attached axles. The motor driver independently controls both DC motors, enabling differential steering. For forward movement, both DC motors run at the same speed and in the same direction. For turning on the spot, the DC motors run at the same speed in opposite directions, causing the robot to pivot around the rear caster wheel. The movement algorithm is explained in Section 1.4.5, Subsystem Integration.

1.4.4 Ball Handling Subsystem

The Ball Handling Subsystem is responsible for collecting and releasing the coloured balls during operation. It consists of two FS90MG servo motors and two 3D-printed clamp arms.¹²

The subsystem operates by the FS90MG servos receiving actuation commands from the Arduino Uno in the Control Subsystem in the form of PWM signals.^{3,12} The PWM pulse width controls servo rotation, thereby opening and closing the clamp arms, which are attached directly to the servos. As explained further in Section 1.4.5, Subsystem Integration, the Arduino sends actuation commands when

the TCRT5000 infrared sensor mounted in the clamp detects a ball and sends a LOW signal to the Arduino.

1.4.5 Subsystem Integration

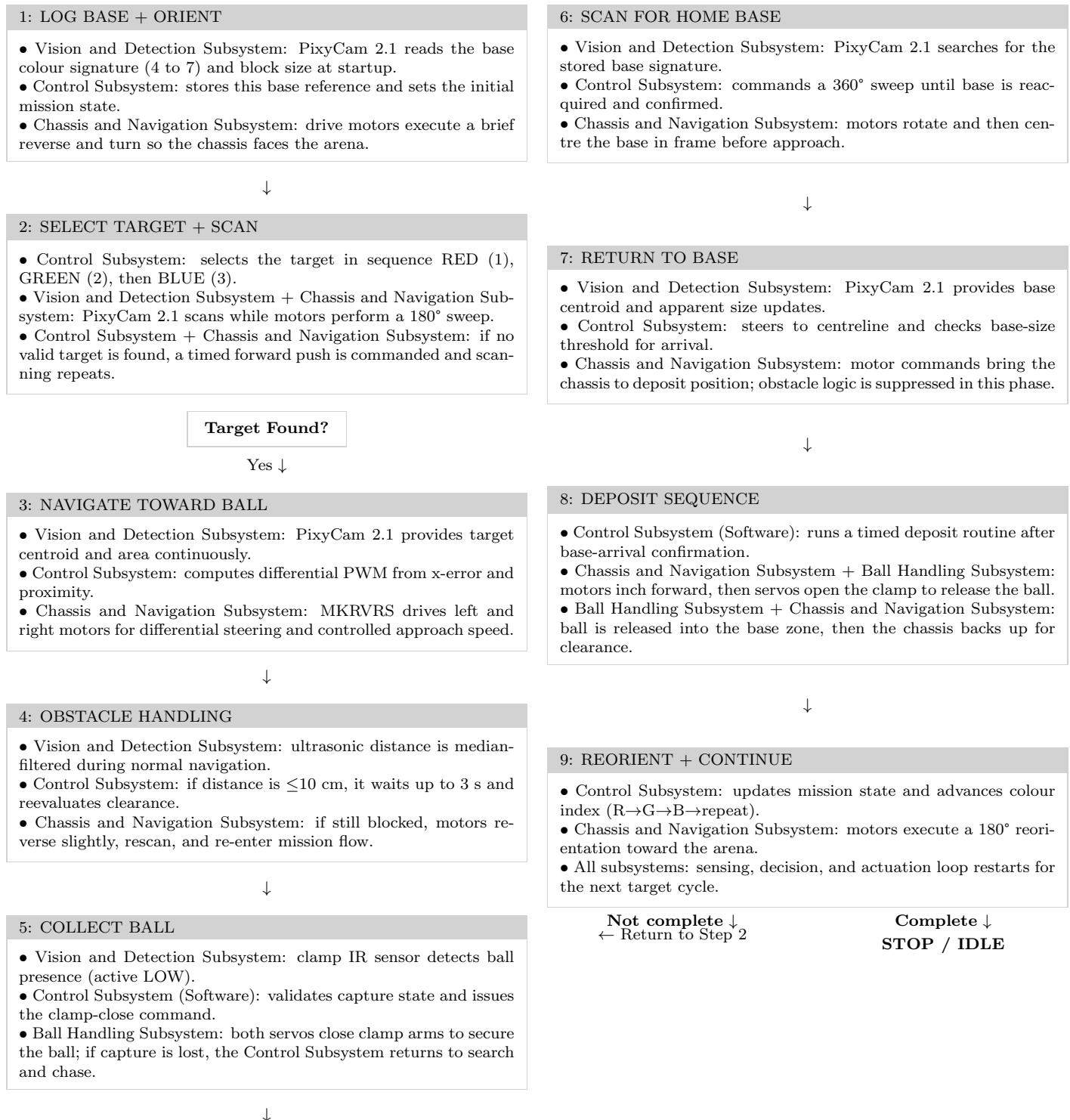


Figure 8: Flowchart of the ball collection and deposit algorithm

1.5 Design Evolution, Testing Influence, and Technical Justification

1.5.1 Mechanical Design Changes

During the development of the final waste sorting robot design, changes were first made to the proposed mechanical design. The initial focus was on adding dedicated component mounts to the chassis. A recessed mount for the rear caster wheel was first added to the bottom plate, as the original configuration made the chassis uneven and caused several components to shift during operation. A dedicated battery mount was then introduced on the bottom plate. This was required as compliance testing showed that the two 9 V batteries were moving during operation. This movement shifted the centre of mass of the robot and resulted in inconsistent steering. The battery mount is shown in Figure 9b.



(a) Technical drawing of the PixyCam and HC-SR04 mount, isometric

(b) Technical drawing of the 9 V battery mount, isometric

Figure 9: Technical drawings of the battery mount and PixyCam and HC-SR04 mount, isometric view.

On the top level, a mount for the LCD display was added. This improved visibility and prevented movement of the display during operation. Following this, a combined mount for the PixyCam 2.1 and HC-SR04 ultrasonic sensor was implemented. This allowed the PixyCam to be positioned at either a 10° or 20° downward angle, improving the detection of nearby balls. It also reduced ambient light interference observed during testing, which affected camera exposure and reduced colour detection accuracy. Furthermore, mounting the ultrasonic sensor on the same bracket saved space and ensured obstacle detection remained aligned with the robot's forward direction. The combined PixyCam and HC-SR04 ultrasonic sensor mount is shown in Figure 9a.

Once these mounting changes were complete, cutouts were introduced in both plates to allow wires to pass cleanly between levels. With the chassis layout established, the front motor mounts were repositioned outward and the width of the front wheels was reduced from 25 mm to 15 mm. This was necessary, as compliance testing showed the wheels were scraping against the chassis during operation. Finally, rubber tyres were added to the wheels due to the inconsistent grip provided by the bare PLA material. The higher coefficient of friction of the rubber improved traction and increased turning accuracy.

1.5.2 Electrical Design Changes

Changes to the proposed electrical design were implemented following modifications to the mechanical design. The solderless breadboard was first replaced with a solderable perfboard. This was necessary, as loose jumper connections may cause intermittent faults during operation. Soldered joints provide fixed and more reliable electrical connections.

After this, the MKRVRS motor driver's regulated 5 V output was connected to the shared main power rails. This was necessary because, during testing, the LCD display was dim under load. Connecting the

regulated 5 V output ensured the LCD display and PixyCam 2.1 received sufficient current, resolving the dimming issue.

Two 1000 μ F capacitors were added next, one across the main power rails and one across the Arduino’s 9 V supply. This was required after components were observed to intermittently turn on and off. This was caused by short voltage dips due to sudden current draw from the N20 geared DC motors and FS90MG servo motors. The capacitors help smooth the power supply by storing charge and releasing it during short voltage dips.¹⁵ A power switch was also added to allow the robot to be powered on and off easily.

Finally, several Arduino Uno pins were reassigned. This was required because, during testing, the HC-SR04 ultrasonic sensor and PixyCam were not working together effectively for navigation. The PixyCam ICSP connector uses the SPI bus, which conflicts with Arduino digital pins 10–13.^{3,5} As such, TRIG was moved from D9 to D8, ECHO from D10 to A2, and the TCRT5000 infrared sensor signal from D11 to D9 (relative to the Design Proposal pin layout). Reassigning these signals freed the SPI lines and allowed both sensors to operate together.

1.5.3 Software Design Changes

From `prefinal_code_v2.ino` to `finaltesting.ino` (see Appendix), the main software change was a shift from simple sequential actions to a more robust state-driven mission flow. The final logic introduced clearer phase control for search, chase, return, and deposit, including orientation state tracking to prevent repeated turns between cycles. Detection handling was also tightened by adding signature confirmation during sweeps and stronger reacquisition behavior when a target was briefly lost, reducing false locks and improving continuity during competition runs.

Obstacle and deposit logic were also revised to improve reliability under real arena conditions. In earlier code, avoidance behavior relied on faster one-step reactions and ultrasonic-based deposit thresholds that were less consistent in practice. The final version uses a staged obstacle routine with timed clearance checks, controlled reverse and rescan behavior, and a calibrated timed deposit sequence with explicit pullback and reorientation. These changes improved repeatability of ball collection cycles, reduced deadlock scenarios near obstacles, and produced more consistent return-to-search transitions after each deposit.

Relative to the Design Proposal flowchart, the final implemented algorithm also simplified and reordered several stages. The proposal separated collection into distinct “collect” and “verify” phases and used a two-phase ultrasonic precision approach for base deposition, whereas the final logic confirms capture directly through IR-triggered clamp closure during chase and performs a fixed-time deposit approach after visual base reacquisition. The final version also added stronger recovery behavior, including periodic 180° sweeps, 360° reacquisition, and orientation state handling between cycles, which produced a more stable mission loop under competition conditions than the original proposal sequence.

1.6 Cost Estimate

The waste sorting robot was constructed primarily from components supplied in the kit, supplemented by additional materials sourced externally.^{3,4,7,10,12,13} Table 1 presents the cost estimate for the robot.

Component	Qty	Supplier	Retail Price (AUD)	Logged Price (AUD)
Arduino Uno	1	Kit	50.00	0.00
Arduino prototyping shield	1	Kit	14.95	0.00
PixyCam 2.1	1	Kit	110.00	0.00
HC-SR04 ultrasonic sensor	1	Kit	1.95	0.00
TCRT5000 infrared sensor	1	Kit	3.50	0.00
N20 DC motor	2	Kit	28.30	0.00
FS90MG servo motor	2	Kit	14.80	0.00
MKR VRS motor driver	1	Kit	6.00	0.00
Rubber tyre	2	Pre-owned	6.10	6.10
Caster wheel	1	Kit	4.15	0.00
Energizer 9 V alkaline battery	2	Woolworths	12.50	12.50
Energizer AAA alkaline battery	4	Woolworths	8.00	8.00
9 V battery connector	2	Kit / Jaycar	3.20	1.60
4×AAA battery case	1	Kit	3.80	0.00
SPDT miniature toggle switch	1	Jaycar	2.95	2.95
16×2 I ² C LCD display	1	Amazon	12.95	12.95
Solderable perfboard	1	Kit	3.10	0.00
1000 μ F capacitor	2	Makerspace	3.90	0.00
Dulux rust paint	$\approx$ 50 g	Bunnings	2.00	2.00
Total			292.15	46.10

Table 1: Cost estimate

Components supplied in the kit were not counted towards the logged price and therefore did not contribute to the 50 AUD budget. The rubber tyres were pre-owned, so their cost was estimated using a listing for a similar tyre on Amazon. The cost of the paint was estimated by taking the proportional cost based on the quantity used by the design team.

2 Final Design Evaluation

2.1 Design Performance and Testing Results

The final waste sorting robot successfully met all of the project’s objectives while abiding by all design restrictions. During the 10-minute time trial on 23 April 2026, the robot collected 16 balls in the operator-specified red, green, and blue order while correctly avoiding the yellow balls. As a result, the design team placed second in the waste sorting competition. This was achieved while remaining within the 200 mm x 200 mm x 200 mm size constraint and the 50 AUD additional materials budget. The robot performed as expected throughout final testing, owing to the several practice runs the team conducted beforehand.

Several aspects of the robot performed well during testing. The ball alignment algorithm consistently aligned the target ball with the clamp mechanism, allowing for accurate collection. It was also particularly effective in recovery scenarios; when the target ball was knocked away by one of the other two robots in the arena, the robot would reverse and recentre on it. Ball colour recognition was similarly reliable, allowing the robot to locate target balls quickly while avoiding the yellow balls. The obstacle detection algorithm also proved integral to the robot’s success. When an obstacle was detected, the robot would wait three seconds, reverse, and perform a 360° sweep to search for nearby balls. If none were detected, it would rotate 180°, move forward, and carry out further sweeps to resume searching. This behaviour allowed the robot to avoid collisions with the arena walls and other robots while quickly returning to ball collection.

The ball searching algorithm, however, did not perform as intended. As explained in Section 1.4.5, Subsystem Integration, when no target ball is detected, the robot is designed to move forward, perform a 180° sweep, and resume forward movement, repeating the process until a target ball is found. During testing, the robot frequently moved forward several times without performing a sweep, causing it to drive across the arena toward the opposite wall without detecting the target ball, as shown in Figure 10. Nevertheless, the impact of this fault was minimised as the obstacle detection algorithm would reroute the robot back toward the ball collection zone before it reached the wall.

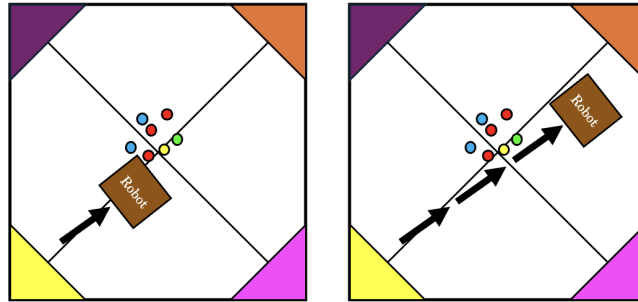


Figure 10: Diagram of the ball searching algorithm fault in Final Testing.

2.2 Challenges and Limitations

During the robot’s development, the design team faced several time-related, resource, and technical challenges. The most significant challenge was the short three-and-a-half-week gap between Compliance Testing and Final Testing. This short gap forced the team to work intensively in the lead-up to allow time for trial runs and troubleshooting. This issue was worsened by a delay in shipping the LCD display, which took over a week to arrive. Because the LCD display needed to be physically measured to design its mount and soldered to the circuit, its late arrival slowed progress on the entire robot. As a result of these time-related challenges, the robot’s operational algorithm was not fully refined by Final Testing, limiting its overall performance.

Another challenge faced by the design team was managing the 50 AUD budget for additional components beyond the provided kit. This constraint required the team to identify cost-effective solutions to issues that could otherwise be resolved with more expensive components. For example, the severity of the power fluctuation issues, addressed using two 1000 μF capacitors, may have been reduced if the budget had allowed for alkaline batteries to be replaced with lithium-polymer batteries. These batteries provide a more stable voltage under load, particularly when components such as N20 DC motors or FS90MG servo motors draw high currents. Nevertheless, the budget constraints did not significantly impact the robot’s performance or its testing results.

The design team also faced various technical challenges while developing the robot. The most notable was managing the 200 mm $\times$ 200 mm $\times$ 200 mm size restriction. This constraint required the team to carefully plan the placement of components, mounts, and wiring cut-outs across both chassis levels to ensure all components fit within the robot without compromising functionality or accessibility for troubleshooting. Subsystem integration also proved difficult. The circuit layout and software had to be repeatedly tested and refined to ensure each subsystem functioned as intended. This process was time-consuming, as some issues spanned multiple subsystems. As a result, individual components often had to be isolated to identify and resolve faults. Despite this, these technical challenges did not limit the robot’s performance or testing outcomes.

2.3 Cost-Benefit Analysis

A simple cost-benefit analysis was carried out on the main components of the robot to determine whether the final design is cost-effective relative to its performance. Each component was rated on a scale of 1 to 5, where 5 is excellent value, 4 good, 3 acceptable, 2 poor, and 1 unjustified. Components supplied in the kit were evaluated using their retail price, as this reflects the true cost of replicating the design. The results are presented in Table 2. Overall, the average rating across the components was 4.29, indicating that the final design is highly cost-effective relative to its performance.

Component	Retail Price (AUD)	Justification	Rating
Arduino Uno	50.00	This component was important as it served as the main control unit of the robot. The price is reasonable given its reliability and broad compatibility.	4
PixyCam 2.1	110.00	This component was critical as it handled all colour detection and target tracking. The high price is justified by its central role in the robot’s performance.	4
HC-SR04 ultrasonic sensor	1.95	This component was important as it enabled obstacle detection during navigation. The price is very low for the functionality provided.	5
TCRT5000 infrared sensor	3.50	This component was important as it detected when a ball was inside the clamp. The price is very low for a reliable and simple sensor.	5
N20 DC motor	28.30	This component was important as it drove the front wheels of the robot. The price is moderate but reasonable for the torque and speed delivered.	4
FS90MG servo motor	14.80	This component was important as it actuated the clamp arms. The price is fair given its durability and reliable performance.	4
MKR VRS motor driver	6.00	This component was important as it controlled the DC motors, allowing the robot to move forward, backward, and turn. The price is low for its functionality.	5
Rubber tyre	6.10	This component was important as it improved wheel traction and turning accuracy. The price is low for a noticeable performance gain.	5
Caster wheel	4.15	This component was important as it provided rear support and allowed the robot to pivot on the spot. The price is very low for its role.	5
Energizer 9 V & AAA alkaline batteries	12.50, 8.00	These components were important as they powered the robot. The price is moderate, but limited capacity and voltage sag under load reduce their value.	3
SPDT miniature toggle switch	2.95	This component was useful as it allowed the robot to be powered on and off easily. The price is very low for the convenience added.	5
16×2 I ² C LCD display	12.95	This component was important as it provided user feedback during operation and helped with troubleshooting. The price is fair for its added clarity and usefulness.	4
1000 μ F capacitor	3.90	This component was important as it smoothed voltage dips caused by the motors and servos. The price is very low for resolving a key issue.	5
Dulux rust paint	2.00	This component was used purely for aesthetic purposes. The price is low, but it contributed nothing to functional performance.	2
Average Rating			4.29

Table 2: Cost-benefit analysis

2.4 Improvements and Future Work

Several changes could be made to improve the robot's performance and extend its real-world applicability. One straightforward improvement would be increasing the PWM duty cycle sent to the N20 geared DC motors, raising the average voltage and wheel speed to allow faster navigation and more collection cycles within the time limit. However, this would require refinements to the movement algorithm, as higher speeds would reduce the robot's ability to stop quickly when an obstacle is detected.

To extend real-world applicability, more substantial changes would be required. First, the clamp mechanism would need redesigning, as its curved arms are specialised for spherical objects of a fixed size; real-world plastic waste varies significantly in shape, size, and rigidity, making a more versatile end-effector such as a soft robotic gripper better suited to irregular objects. Second, the current 9 V alkaline and AAA batteries lack sufficient capacity for extended real-world operation, so upgrading to a rechargeable lithium-polymer battery pack would improve runtime while reducing long-term costs and battery waste.¹⁴

2.5 Lessons Learned

Throughout the project, the design team developed a wide range of skills. On the technical side, the team learnt the basics of CAD modelling, electrical circuit design, soldering, and coding, all of which were essential to building the robot. On the non-technical side, the team developed strong project management and teamwork skills. Across the ten-week project, each member learnt how to be punctual, plan meetings, and communicate effectively in a semi-professional manner. A key outcome of the project was the design team's improved ability in project planning and time management. This was particularly important in the waste sorting robot project, as tight deadlines and simultaneous subsystem development required careful scheduling. The team managed this using a Gantt chart. Overall, all these technical and non-technical skills can be applied to both future individual and group projects.

3 Conclusion

Overall, the final waste sorting robot successfully met the project's performance objectives, autonomously sorting red, green, and blue balls representing HDPE, PET, and PP plastic types while avoiding yellow balls representing non-recyclable material, within the defined arena and in an operator-specified order. This was achieved while remaining within the 200 mm x 200 mm x 200 mm size constraint and the 50 AUD additional materials budget.

The final design is a self-contained, WALL-E inspired system divided into four subsystems. The robot is built around a dual-level PLA chassis with dedicated component mounts, cutouts for cables, and a detachable outer shell. It is driven by two front wheels and a rear caster wheel, allowing it to turn on the spot. The robot is controlled by an Arduino Uno, powered by two 9 V alkaline batteries and four AAA batteries, and uses several sensors to detect balls and obstacles, which it collects with a curved, servo-actuated clamp. The design is also highly cost-effective relative to its performance.

During final testing, the robot performed well, collecting 16 balls during the 10-minute time trial and placing the design team second in the waste sorting competition. This success demonstrates that autonomous robots can be applied effectively to plastic waste sorting, offering a way to improve recycling efficiency and sustainability on university campuses and in public spaces. Nevertheless, the current final design is highly specialised for ball collection and would require significant modification to be suitable for real-world plastic waste sorting. Ultimately, the project was a success, providing valuable insight into the application of automated systems in waste sorting.

4 References

- [1] Cleanaway Waste Management Ltd., *Recycling Behaviours Report 2024: Australians and Recycling: Attitudes, Behaviours and Outlook*, May 2024. [Online]. Available: https://cleanaway2stor.blob.core.windows.net/cleanaway2-blob-container/2024/05/CWY_RBR2024_Report_7MAY2024-FINAL.pdf. [Accessed: May 1, 2026].
- [2] United Nations Environment Programme, “Plastic pollution,” UNEP. [Online]. Available: <https://www.unep.org/plastic-pollution>. [Accessed: May 1, 2026].
- [3] Arduino, “UNO R3,” *Arduino Documentation*. [Online]. Available: <https://docs.arduino.cc/hardware/uno-rev3/>. [Accessed: May 1, 2026].
- [4] PixyCam, “Color Connected Components,” *PixyCam 2.1 Documentation*. [Online]. Available: https://docs.pixycam.com/wiki/doku.php?id=wiki:v2:color_connected_components. [Accessed: May 1, 2026].
- [5] PixyCam, “Color Connected Components API,” *PixyCam 2.1 Documentation*. [Online]. Available: https://docs.pixycam.com/wiki/doku.php?id=wiki:v2:ccc_api. [Accessed: May 1, 2026].
- [6] Arduino, “analogWrite(),” *Arduino Documentation*, May 9, 2025. [Online]. Available: <https://docs.arduino.cc/language-reference/en/functions/analog-io/analogWrite/>. [Accessed: May 1, 2026].
- [7] ElecFreaks, *Ultrasonic Ranging Module HC-SR04 Datasheet*. [Online]. Available: <https://cdn.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf>. [Accessed: May 1, 2026].
- [8] D. Nedelkovski, “Ultrasonic Sensor HC-SR04 and Arduino – Complete Guide,” *How To Mechatronics*. [Online]. Available: <https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/>. [Accessed: May 1, 2026].
- [9] DIY Machines, “HC-SR04 Ultrasonic Sensor – What it is and how to use it,” *YouTube*. [Online]. Available: <https://www.youtube.com/watch?v=1E4TDZC-xaY&t=101>. [Accessed: May 1, 2026].
- [10] Vishay Intertechnology, *TCRT5000 Reflective Optical Sensor with Transistor Output Datasheet*. [Online]. Available: <https://www.vishay.com/docs/83760/tcrt5000.pdf>. [Accessed: May 1, 2026].
- [11] Instructables, “TCRT5000 Infrared Reflective Sensor – How It Works and Example Circuit With Code.” [Online]. Available: <https://www.instructables.com/TCRT5000-Infrared-Reflective-Sensor-How-It-Works-a/>. [Accessed: May 1, 2026].
- [12] FEETECH, *FS90MG Micro 1.8 kg.cm Metal Gears Analog Servo Datasheet*. [Online]. Available: <https://www.gottron.be/media/files/Downloads/FS90MG.pdf>. [Accessed: May 1, 2026].
- [13] Phipps Electronics, “1602 16x2 HD44780 LCD Display with Soldered I2C Module – Support Documentation,” May 10, 2025. [Online]. Available: <https://www.phippselectronics.com/support/1602-16x2-hd44780-lcd-display-with-soldered-i2c-module-support-documentation/>. [Accessed: May 1, 2026].
- [14] Adafruit, “Voltages,” *Li-Ion & LiPoly Batteries*, Jul. 29, 2012, revised Jul. 8, 2025. [Online]. Available: <https://learn.adafruit.com/li-ion-and-lipoly-batteries/voltages>. [Accessed: May 1, 2026].

- [15] Electronics Tutorials, “Capacitor Charge and Discharge.” [Online]. Available: https://www.electronics-tutorials.ws/capacitor/cap_2.html. [Accessed: May 1, 2026].

5 Appendices

5.1 Source Code Links

Draft pre-final code (`prefinal_code_v2.ino`): [View on Google Docs](#)

Final testing code (`FINALTESTING.ino`): [View on Google Docs](#)